

Problem

Design a passive band-stop filter with $\omega_0 = 10$ rad/s and $Q = 20$.

Step-by-step solution

Step 1 of 6

Consider a passive bandstop filter with quality factor $Q = 20$ and center frequency $\omega_0 = 10$ rad/s.

Draw the following passive bandstop filter circuit diagram:

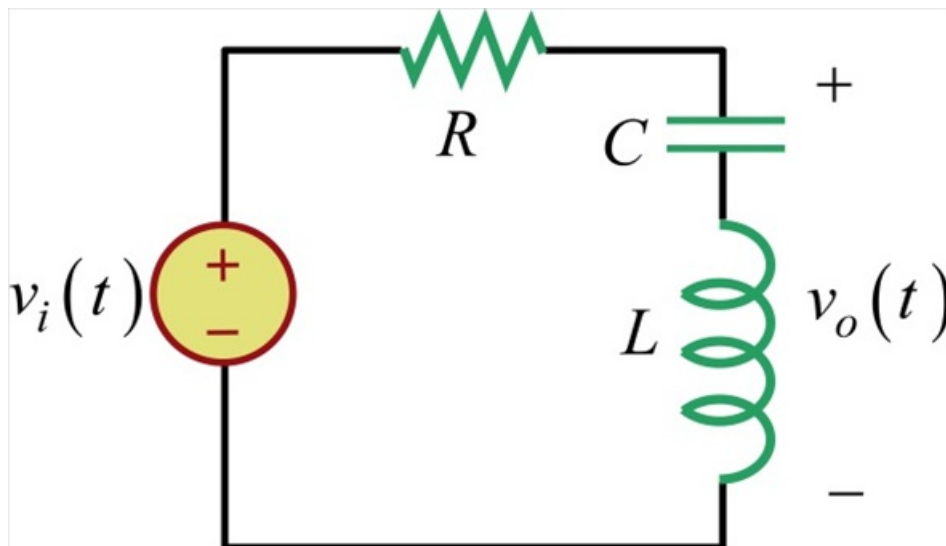


Figure 1: Passive bandstop filter.

Step 2 of 6

Write the expression for the frequency of rejection ω_0 .

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad \dots\dots (1)$$

Substitute 10 rad/s for ω_0 in this expression.

$$10 = \frac{1}{\sqrt{LC}}$$

$$LC = 0.01 \quad \dots\dots (2)$$

Step 3 of 6

Write the expression for the quality factor Q .

$$Q = \frac{\omega_0 L}{R}$$

$$R = \frac{\omega_0 L}{Q}$$

Substitute 20 for Q , 10 rad/s for ω_0 in this expression.

$$\begin{aligned} R &= \frac{10L}{20} \\ &= 0.5L \end{aligned}$$

Let the value of inductor L to be 10 H.

$$\begin{aligned} R &= 0.5(10) \\ &= 5 \end{aligned}$$

Therefore, the value of the resistor R is 5 Ω .

Step 4 of 6

Calculate the value of capacitor C .

Substitute 10 H for L in this expression 2.

$$(10)C = 0.01$$

$$C = 1 \text{ mF}$$

Therefore, the value of capacitor C is 1 mF.

Step 5 of 6

Redraw the Figure 1 as shown in Figure 2.

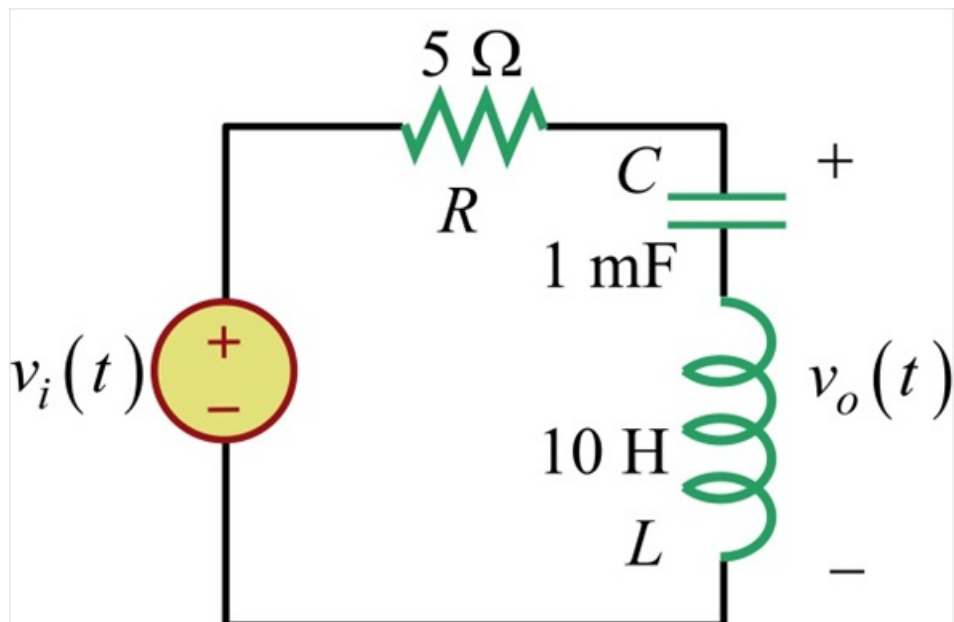


Figure 2

Step 6 of 6

Hence, the designed circuit is as shown in [Figure 2](#).